

Multiple Choice (10 marks)

1. Say the pupil of your eye has a diameter of 5 mm and you have a telescope with an aperture of 50 cm. How much more light can the telescope gather than your eye?
 - (a) 10000 times more
 - (b) 100 times more
 - (c) 1000 times more
 - (d) 10 times more
2. What is true about the rotation of a solid disk?
 - (a) The velocity is the same on every point on the disk
 - (b) The part closer to the axis has a longer period
 - (c) The period of rotation is the same on every point on the disk
 - (d) The part closer to the axis has a shorter period
3. The resolution of a telescope is a measure of its:
 - (a) Ability to magnify distant objects
 - (b) Ability to measure the angular separation of objects
 - (c) Ability to measure the distance between objects
 - (d) Light-collecting efficiency
4. A comet of mass m impacts the earth (mass M radius R) at the minimum impact speed. What is the expression for the total energy released in the impact?
 - (a) $m \cdot v$
 - (b) $0.5 \cdot m / (R^3)$
 - (c) $0.5 \cdot m \cdot (2 GM/R)$
 - (d) $0.6 \cdot G(M^2)/R$
5. Find the best approximation for the surface temperature of the Sun:
 - (a) 6000 K
 - (b) 7000 K
 - (c) 9000 K
 - (d) 13000 K

True / False (10 marks)

Answer True or False

6. True or False: The atmosphere of Mars is primarily composed of carbon dioxide (CO₂), making it unsuitable for human respiration.
7. True or False: The sky appears blue because the atmosphere preferentially scatters shorter wavelengths of light, such as blue.
8. True or False: Mars experiences more extreme seasons due to its more eccentric orbit compared to Earth's nearly circular orbit.
9. True or False: The ecliptic refers to the axial tilt of the Earth, which influences the seasons throughout the year.
10. True or False: The Cassini division is a large gap in Saturn's rings created by gravitational interactions with the moon Mimas.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the various surface processes evident on Venus, analyzing their implications for the planet's geological history and current geological activity.
12. Discuss the relationship between a Mars year and an Earth year, analyzing the implications of this difference on potential human exploration and colonization of Mars.

End of Paper